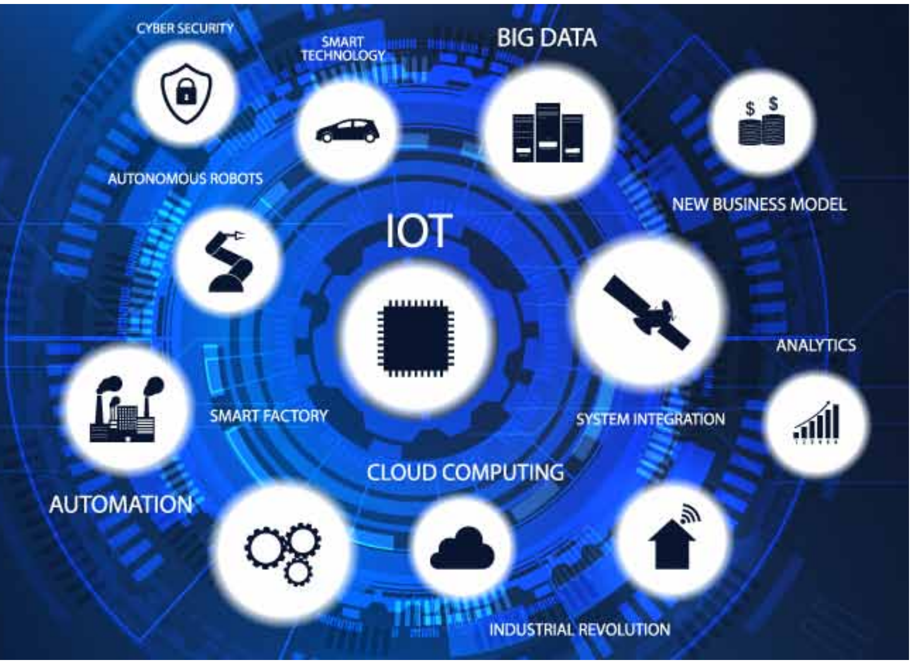
Industry 4.0 and Embedded Internet of Things: An Introduction

The fourth industrial revolution, commonly referred to as Industry 4.0, is all about making businesses smarter. It is focused on deepening the impact of digital technologies by making machines more autonomous and enabling them to communicate with one another. These machines can now manage enormous amounts of data in ways that humans simply cannot handle. This article introduces you to Embedded Internet of Things (EIoT) and Industry 4.0.



The third industrial revolution was focused on converting mechanical and analogue processes to digital ones to be ready for advanced automation. Similar to the transition from steam to electricity in the second industrial revolution, the advent of Industry 4.0 technology marks a major transformation in how companies conduct their operations.

The effects of Industry 4.0 on production are extensive. It is being used to increase operational effectiveness, improve demand forecasting, eliminate data silos, engage in predictive maintenance, provide virtual training to employees, and more. Industry 4.0 encompasses manufacturing from planning to delivery as part of a larger concept known as digital transformation. It includes solutions for deep analytics, shop floor data sensors, smart warehouses, simulated modifications, as well as product and asset tracking.

Industry 4.0 technologies assist manufacturers in bridging the gap between previously distinct processes and a more transparent, viewable perspective across the entire business with a wealth of useful insights. The emergence of digital solutions and cutting-edge technology, which are frequently linked to Industry 4.0, have made this convergence possible.

Some of the key technologies associated with Industry 4.0 are:

- Industrial Internet of Things
- Smart wearable gadgets
- Smart cities and advanced monitoring
- Advanced robotics
- Digital twins
- Big Data analytics
- E-governance
- Additive manufacturing (AM)
- Cloud computing
- Augmented reality (AR) and virtual reality (VR)
- Quantum computing based systems
- Simulation platforms and suites

By integrating formerly disjointed systems and processes with